Low-Voltage Bandgap Reference Circuit in 28nm CMOS

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Abstract—**This paper presents a hybrid adjusted temperature compensation circuit for reducing the temperature drift of the bandgap reference. Combining first-order bandgap current, nonlinear compensation current, and temperature curvature compensation current together, a temperature insensitive reference voltage can be obtained in proposed circuit. Designed and verified in UMC 28nm CMOS technology with Cadence IC615, the proposed circuit achieves a post-layout simulation temperature drift of 5.48 ppm/°C in the range of -20°C to 120°C with a supply voltage of 1.05-V.**

Keywords-bandgap reference; temperature compensation; low-voltage

I. INTRODUCTION

Bandgap voltage reference is a critical block for analog devices or mixed-systems [1]. It is widely used in LDO, ADC/DAC, PLL, DRAM, etc. The temperature stability and accuracy of the reference voltage can directly affect the performance of the entire system.

In the design process of the traditional bandgap reference, a single first-order compensation is performed on the V_{EB} (emitter-base voltage of the PNP bipolar transistor). The temperature coefficient is generally tens ppm/°C, which is difficult to satisfy the high-precision ADC, PLL, etc. In recent years, researchers have proposed several effective high-order curvature compensation techniques [4-7]. However, these papers only adopt single curvature correction or compensation for the high-order nonlinear components of the V_{EB}. They have large temperature coefficient and high supply voltage, and which are difficult to apply in low-voltage nanoscale circuits.

The advanced 28nm CMOS technology is difficult for the analog circuit design, especially the second-order effect of the device proposes more challenges for the design of the bandgap reference. This paper presents a hybrid adjusted temperature compensation bandgap reference circuit and can achieve an improved simulated performance.

The remainder of this paper is organized as follows, section II describes the basic working principle of CMOS bandgap reference. In section III, we present a hybrid adjusted temperature compensation topology of bandgap reference. In section IV, we show the simulation results along with related discussions. The conclusions are drawn in the last section.

II. BASIC BANDGAP DESIGN PRINCIPLE

Widlar firstly proposed the concept of bandgap reference in [2]. The V_{BE} has a negative temperature coefficient (approximately -1.85 mV/°C), while the thermal voltage V_T has a positive temperature coefficient (approximately 0.086 mV/°C). If mentioned voltages with opposite temperature characteristics can be added together with appropriate ratio, the resulting voltage can be approximated with a zero temperature coefficient. The mentioned principle has been illustrated in Fig. 1, the output voltage can be expressed as

$$V_{ref} = V_{BE} + KV_T \tag{1}$$

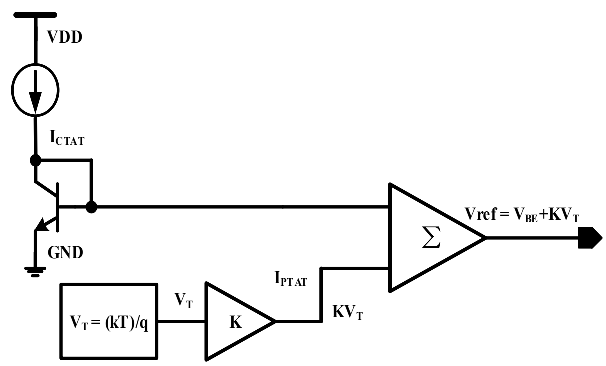


Figure 1. Basic principle of bandgap reference.

III. THE PROPOSED BANDGAP REFERENCE CIRCUIT

In this paper, a bandgap reference with hybrid adjusted temperature compensation circuit was designed to achieve a small temperature coefficient reference voltage in a wide temperature range. As shown in Fig. 2, the hybrid adjusted temperature compensation circuit includes a startup circuit, a low-voltage bandgap reference circuit, a nonlinear compensation circuit, a temperature curvature compensation circuit, and an output stage circuit.

In the low-voltage bandgap reference circuit, the operation currents I_{PTAT} and I_{CTAT} are given by

$$I_{PTAT} = \frac{V_{EB1} - V_{EB2}}{R1} = \frac{V_T \ln(N)}{R1} = \frac{KT \ln(N)}{qR1} \tag{2}$$

$$I_{CTAT} = I_{R2} = \frac{V_{EB1}}{R2} \tag{3}$$

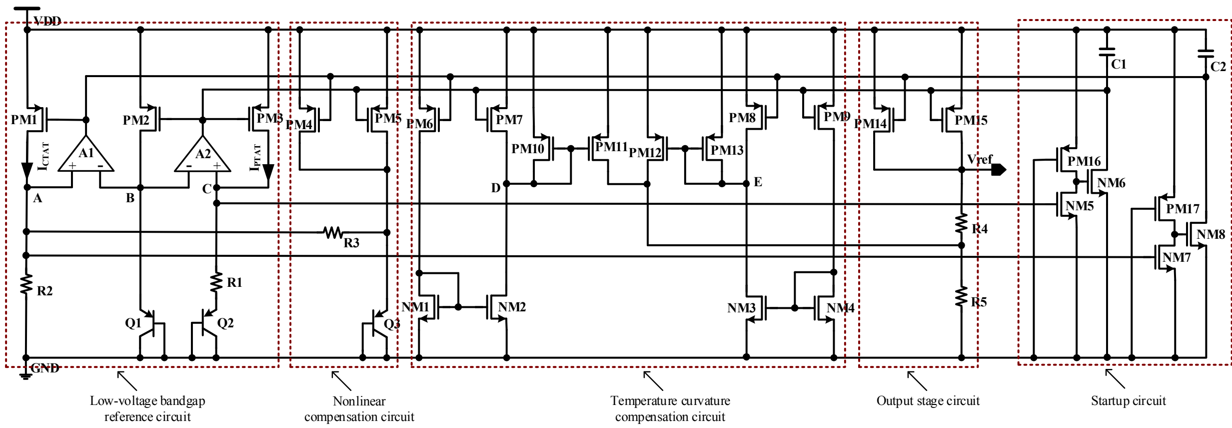


Figure 2. Proposed bandgap reference circuit.

where the V_T is the thermal voltage, k is the Boltzmann constant (1.38×10^{-23} J/K), q is the electron charge (1.6×10^{-19} C), T is the absolute temperature, and N is the ratio of emitter-area between Q2 and Q1.

Superimposing the current of PM14 and PM15, the output voltage Vref across resistors R4 and R5 can be expressed as

$$Vref = (R4+R5)\left(\frac{V_{EB1}}{R2}+\frac{KT\ln(N)}{qR1}\right) = K_1 V_{EB1} + K_2 T \tag{4}$$

where $K_1 = (R4+R5)/R2$, $K_2 = K(R4+R5)\ln(N)/qR1$. When $\partial Vref/\partial T = 0$, in other words, $K_1 \, \partial V_{EB1}/\partial T = K_2$, the output voltage Vref is not related to temperature.

The V_{EB} (emitter-base voltage of the PNP bipolar transistor) is not completely a negative linear relationship with temperature. We have the following equation [3]

$$V_{EB}(\mathrm{T}) = V_{G0} - \left[V_{G0} - V_{EB}(\mathrm{T}_0)\right]\frac{T}{T_0} - (\eta-\alpha)\frac{kT}{q}\ln\frac{T}{T_0} \tag{5}$$

where V_{G0} is the extrapolated bandgap voltage with dependence on temperature (1.206 V @ 0 K), η is a constant depending on the technology and α is the order of the temperature dependence of the collector current[4].

The collector current of Q3 is approximately independent of the temperature then from (5), α = 0. Therefore we obtain the following expression for V_{EB3}

$$V_{EB3}(\mathrm{T}) = V_{G0} - \left[V_{G0} - V_{EB3}(\mathrm{T}_0)\right]\frac{T}{T_0} - \eta\frac{kT}{q}\ln\frac{T}{T_0} \tag{6}$$

The collector current of Q1 is a negative relationship with temperature then from (5), α = -1. Therefore we get the following expression for V_{EB1}

$$V_{EB1}(\mathrm{T}) = V_{G0} - \left[V_{G0} - V_{EB1}(\mathrm{T}_0)\right]\frac{T}{T_0} - (\eta+1)\frac{kT}{q}\ln\frac{T}{T_0} \tag{7}$$

Therefore, the current I_{R3} across the resistor R3 can be given by

$$I_{R3} = \frac{V_{EB3}-V_A}{R3} = \frac{V_{EB3}-V_{EB1}}{R3} = \frac{V_T\dfrac{\ln(T)}{T_0}}{R3} = \frac{V_{NL}}{R3} \tag{8}$$

where V_{NL} is the nonlinear voltage term. Using a Taylor series expansion in (8) it can be rewritten in the form of

$$V_{NL} = \alpha_0 + \alpha_1 T + \alpha_2 T^2 + \alpha_3 T^3 + \cdots + \alpha_n T^n \tag{9}$$

where α_0 is a constant term, $\alpha_1 T$ is a linear term and the $\alpha_2 T^2 + \alpha_3 T^3 + \cdots + \alpha_n T^n$ are the nonlinear terms[4].

The output voltage Vref can be revised as

$$Vref = (R4+R5)\left[\left(\frac{V_{EB1}}{R2}+\frac{V_T\ln(N)}{R1}\right)-\left(\frac{V_{NL}}{R3}\right)\right] \tag{10}$$

In the temperature curvature compensation circuit, the size of NM2 is A_1 times that of NM1, PM11 is A_2 times as great as PM10, NM3 is B_1 times larger than NM4, and PM12 is B_2 times as great as PM13. At node D, when the drain current of NM2 is smaller than PM7, D is high level. Transistors PM10 and PM11 are off and no current flows into transistor PM11. However, when the drain current of NM2 is larger than PM7, D is low potential. Therefore, transistors PM10 and PM11 are turned on and the compensation current flows into resistor R5. Similarly, at node E, when the drain current of NM3 is smaller than PM8, E is high level. Then, transistors PM12 and PM13 are off and the proposed temperature curvature compensation circuit does not generate any current or voltage to the output stage circuit. Conversely, when the drain current of NM3 is larger than PM8, E is low potential. Thus, transistors PM12 and PM13 are turned on and the compensation current flows into resistor R5.

When the temperature T is smaller than T_1, low-temperature compensation is performing. Conversely, when the temperature T is greater than or equal to T_1, low temperature compensation is not working. Thus, the drain current of PM11 can be given by

$$I_{PM11,D} = \begin{cases} A_2\left(A_1 \bullet I_{CTAT} - I_{PTAT}\right) & T < T_1 \\ 0 & T \geq T_1 \end{cases} \tag{11}$$

When the temperature T is larger than T2, high-temperature compensation is implementing. However, when the temperature T is less than or equal to T2, high temperature compensation is not acting [7]. Therefore, the drain current of PM12 can be expressed as

$$
Vref=\begin{cases}(R4+R5)\left[\left(\dfrac{V_{EB1}}{R2}+\dfrac{V_T\ln(N)}{R1}\right)-\left(\dfrac{V_{NL}}{R3}\right)\right]+A_2\left(A_1I_{CTAT}-I_{PTAT}\right)R5 & T<T_1\\ (R4+R5)\left[\left(\dfrac{V_{EB1}}{R2}+\dfrac{V_T\ln(N)}{R1}\right)-\left(\dfrac{V_{NL}}{R3}\right)\right] & T_1\le T\le T_2\\ (R4+R5)\left[\left(\dfrac{V_{EB1}}{R2}+\dfrac{V_T\ln(N)}{R1}\right)-\left(\dfrac{V_{NL}}{R3}\right)\right]+B_2\left(B_1I_{PTAT}-I_{CTAT}\right)R5 & T>T_2\end{cases} \quad (13)
$$

$$
I_{PM12,D}=\begin{cases}B_2\left(B_1I_{PTAT}-I_{CTAT}\right) & T>T_2\\ 0 & T\le T_2\end{cases} \quad (12)
$$

At present, the output voltage Vref can be expressed by (13). Where I_{PTAT} and I_{CTAT} can be obtained from (2) and (3). By reasonably adjusting the scales of R1, R2, R3, R4, R5, A_1, A_2, B_1, and B_2, the small temperature drift of output voltage Vref can be acquired.

IV. Simulation Results

The hybrid adjusted temperature compensation bandgap reference was designed and verified in UMC 28 nm CMOS process with Cadence IC615. The simulation results of the output voltage with temperature in the range from -20°C to 120°C are shown in Fig. 3. The temperature coefficients of pre-simulation and post-layout simulation are 4.14 ppm/°C and 5.48 ppm/°C, respectively.

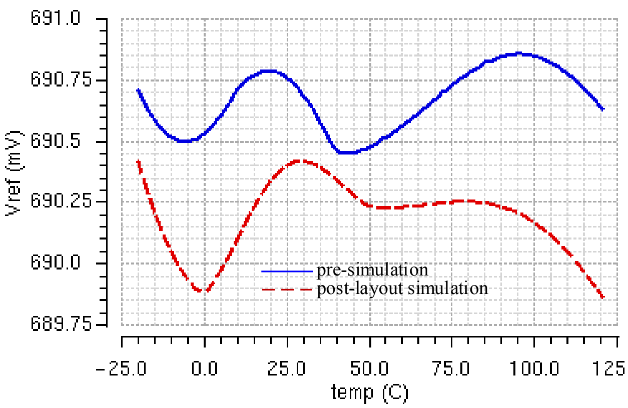


Figure 3. Temperature dependence of proposed circuit.

When the supply voltage is a step signal with a rise time of 10μs, the transient simulation is performed on the proposed circuit. The simulation result is shown in Fig. 4 (a), output voltage is stable at 12μs. Fig. 4 (b) shows the power supply rejection ratio (PSRR) of this bandgap reference and the PSRR is approximately 55.4dB.

To investigate the reliability and assess yield on proposed bandgap reference, Monte Carlo simulations (100 runs) were run and the simulation results are shown in Fig. 5 [8]. The Fig. 5 (a) shows that the reference voltage varied from 689.73 mV to 691.85 mV under a worst case. Meanwhile, Fig. 5 (b) shows the histogram of the calculated temperature coefficient (TC) over 100 simulation runs for the proposed circuit. Across 100 runs, 88% of samples show TC < 8 ppm/°C.

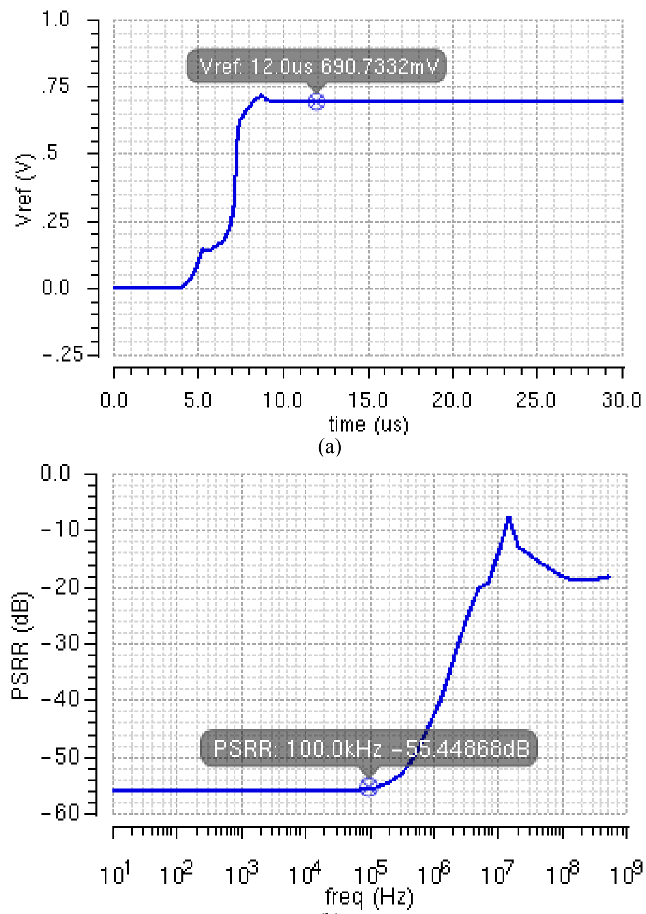
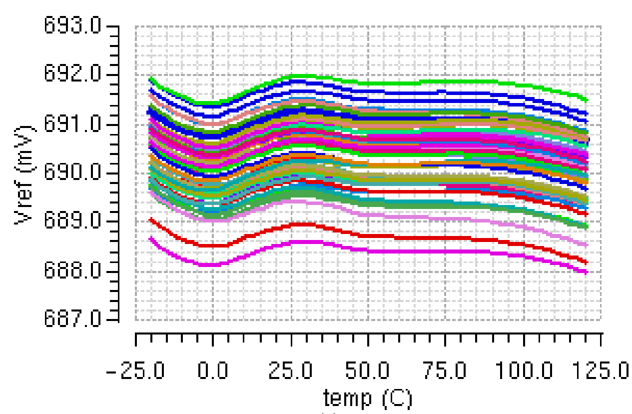
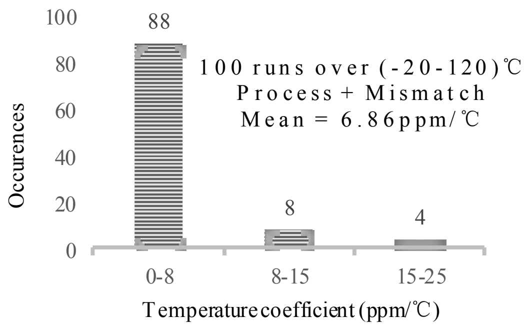


Figure 4. (a) Transient simulation, and (b) PSRR simulation.



(a)



(b)

Figure 5. (a) Monte Carlo simulation results with Process + Mismatch, and (b) the histogram of the calculated TC from (a).

The layout of proposed bandgap reference is shown in Fig. 6, which occupies an active area of 0.018mm^2. A comparison between the proposed circuit and prior designs is shown in Table I. It can be seen that the proposed bandgap reference obtains temperature coefficient of 5.48 ppm/°C outperforms the designs of [6], [9], and [10] over a wide temperature range.

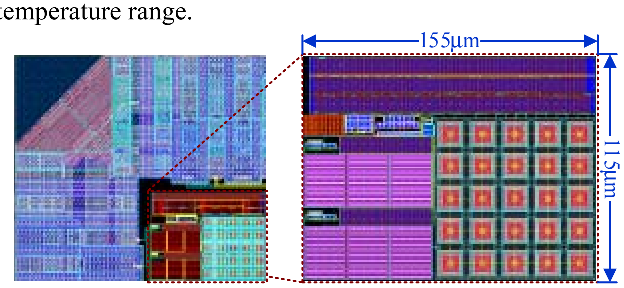


Figure 6. Layout of proposed bandgap reference.

TABLE I. PERFORMANCE COMPARISON OF PROPOSED CIRCUIT WITH PRIOR DESIGNS

	Process	Supply Voltage	Output Voltage	TC/Temp Range
This work simulated	28 nm CMOS	1.05 V	0.69 V	5.48 ppm/°C/(-20-120)°C
[6] measured	28 nm CMOS	1.4 V	0.6 V	33 ppm/°C/(-40-125)°C
[9] simulated	0.5 µm CMOS	3.3 V	1.25 V	11.6 ppm/°C/(-40-88)°C
[10] measured	90 nm CMOS	1.6 V	0.77 V	39.5 ppm/°C/(-40-125)°C

V. CONCLUSIONS

A hybrid adjusted temperature compensation bandgap reference circuit is presented in the paper. The proposed topology achieves an improved simulated performance in comparison to prior designs. Designed and verified in UMC 28nm CMOS process with Cadence IC615, the proposed topology achieves a post-layout simulation temperature coefficient of 5.48 ppm/°C in the range of -20°C to 120°C with 1.05-V supply voltage.

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